

[1.7] Chinese Remainder Theorem.

Def 1.7.1 For ideals I, J in a ring R , we define

One can see that $I+J, I \cap J, I \cdot J$ are ideals, we need this.

Also observe $I \cdot J \subseteq I \cap J \subseteq I+J$.

Min example: $2\mathbb{Z} \cap 4\mathbb{Z} = 4\mathbb{Z}, 2\mathbb{Z} \cdot 4\mathbb{Z} = 8\mathbb{Z},$
 $4\mathbb{Z} + 6\mathbb{Z} = \gcd(4,6)\mathbb{Z} = 2\mathbb{Z}.$

Rem 1.7.2 Consider the case $I = \langle f_1, \dots, f_s \rangle$ and $J = \langle g_1, \dots, g_t \rangle$ with $f_1, \dots, f_s, g_1, \dots, g_t \in k[x_1, \dots, x_n]$.
 $I+J = \langle f_1, \dots, f_s, g_1, \dots, g_t \rangle$, the varieties are $V(\underbrace{f_1, \dots, f_s}_I) \cap V(\underbrace{g_1, \dots, g_t}_J) = V(\underbrace{f_1, \dots, f_s, g_1, \dots, g_t}_{I+J})$

$I \cdot J = \langle f_i g_j : i=1, \dots, s, j=1, \dots, t \rangle$. This corresponds to
 $V(f_1, \dots, f_s) \cup V(g_1, \dots, g_t) = V(f_i g_j : i=1, \dots, s, j=1, \dots, t)$

$$I+J = \{f+g : f \in I, g \in J\}$$

$I \cap J$ already defined, just the intersection

$$I \cdot J = \left\{ \sum_{i=1}^m f_i \cdot g_i : m \in \mathbb{N}, f_i \in I, g_i \in J \right\}$$

$$I = \{a_1, a_2, a_3, a_4, \dots\}$$

$$J = \{b_1, b_2, b_3, b_4, \dots\}$$

$$a_2 b_3 \quad a_1 b_4 \quad a_4 b_2$$

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Def 1.7.3. Ideals I and J in a ring R are called coprime if $I+J=R$.

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Rem. 1.7.4. Let's consider $\mathbb{Z}$ and $k[x]$

Every ideal in $\mathbb{Z}$ has the form $m\mathbb{Z}$ with $m \in \mathbb{N}$. Two such ideals $m\mathbb{Z}$ and $n\mathbb{Z}$ ($m, n \in \mathbb{N}$) are coprime iff $m\mathbb{Z} + n\mathbb{Z} = \mathbb{Z}$, which means $\gcd(m, n) = 1$ (so, m and n are coprime).

It's just the same for $k[x]$.

Def 1.7.5. For rings R and S , we define the product ring of R and S as the Cartesian product $R \times S = \{(r, s) : r \in R, s \in S\}$ and the operations $(r_1, s_1) + (r_2, s_2) = (r_1 + r_2, s_1 + s_2)$ and $(r_1, s_1) \cdot (r_2, s_2) = (r_1 \cdot r_2, s_1 \cdot s_2)$. This is in fact a ring with $0_{R \times S} = (0_R, 0_S)$ and $1_{R \times S} = (1_R, 1_S)$.

Theorem 1.7.6 (Chinese Remainder Theorem = CRT, the version $R \rightarrow R/I \times R/J$)

Let I and J be coprime ideals in a ring R . Then the map

$$\varphi: R \rightarrow R/I \times R/J, \quad \varphi(r) = ([r]_I, [r]_J)$$

is a surjective ring homomorphism.



Proof: Clearly, ϕ is a homomorphism. We need to show it is surjective.

Consider an arbitrary element of $R/I \times R/J$, say $([a]_I, [b]_J)$, with $a, b \in R$.
We need to find an $r \in R$ with $\phi(r) = ([a]_I, [b]_J)$, that is, with

$$\begin{aligned} r &\equiv a \pmod{I} \\ r &\equiv b \pmod{J} \end{aligned}$$

I and J are coprime, i.e., $I + J = R \Rightarrow u + v = 1$ for some $u \in I$ and some $v \in J$.

	mod I	mod J
$v \equiv$	1	0
$u \equiv$	0	1

$\Leftarrow$

$$\begin{array}{l} v = 1 - u \\ u \in I \\ \hline u = 1 - v \\ v \in J \end{array}$$

$\Downarrow$
 $r = av + bu$ is the choice, since

$$r \equiv a \cdot 1 + b \cdot 0 \pmod{I}$$

$$r \equiv a \cdot 0 + b \cdot 1 \pmod{J}. \quad \square$$

Question: Why do things fail without coprimality assumption?
 $2\mathbb{Z}$ and $4\mathbb{Z}$ not coprime.

$$\mathbb{Z}/2\mathbb{Z} = \{[0]_2, [1]_2\}$$

$$\mathbb{Z}/4\mathbb{Z} = \{[0]_4, [1]_4, [2]_4, [3]_4\}$$

There is no $r \in \mathbb{Z}$ such that

$$r \equiv 0 \pmod{2\mathbb{Z}}$$

$$r \equiv 1 \pmod{4\mathbb{Z}}$$

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1.7.7. Extended Euclidean Algorithm. For the proof to be turned into an algorithm, we need to be able to calculate $u \in I$ and $v \in J$ such that $u+v=1$. For example, we may have $R=\mathbb{Z}$ and $I=m\mathbb{Z}$, $J=n\mathbb{Z}$ with $\gcd(m,n)=1$. So we need to find a representation of 1 as $1 = \underbrace{m \cdot c}_u + \underbrace{n \cdot d}_v$ where $c, d \in \mathbb{Z}$. The constants c and d are called Bézout

coefficients. They can be found by the extended Euclidean algorithm. We already know the Euclidean Algorithm to find $\gcd(m,n)$. Now we want to see according to which idea we can extend it.

Example: $\gcd(19, 5) =$ Let's attach expressions to integer values starting with

$$\begin{aligned} \gcd(19, 5) &= \\ \gcd(19-3 \cdot 5, 5) &= \\ \gcd(4, 5) &= \end{aligned}$$

$$\begin{aligned} \gcd(4, 5-4) &= \\ \gcd(4, 1) &= \\ \gcd(4-4 \cdot 1, 1) &= \\ \gcd(0, 1) &= \end{aligned}$$

$= 1$

$$\underbrace{m=19}_{m=19} - 3 \cdot \underbrace{n=5}_{n=5} = \underbrace{(m-3 \cdot n)=4}_{m-3n=4}$$

$$\underbrace{m-3n=4}_{m-3n=4} \quad \underbrace{n=5}_{n=5}$$

$$4n - m = 5 - 4$$

and so on.

With linear algebra notation

$$\begin{array}{l} m=19 \\ n=5 \end{array} \quad \left[\begin{array}{cc|c} 1 & 0 & 19 \\ 0 & 1 & 5 \end{array} \right] \rightsquigarrow$$

transform this system until you get $\left[\begin{array}{cc|c} * & * & 0 \\ c & d & 1 \end{array} \right]$

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Implementation in Sage Math

def ext_eucl(m, n):

f = vector([m, 1, 0]) # f[0] == f[1] * m + f[2] * n

g = vector([n, 0, 1]) # g[0] == g[1] * m + g[2] * n

while g[0] != 0:

q = f[0] // g[0] # quotient of long division

f, g = g, f - q * g

return f

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1.7.8. "Making" homomorphisms injective. Assume we have a ring homomorphism $\varphi: R \rightarrow S$ and assume that $\ker(\varphi) \neq \{0\}$, that means, φ is not injective, but we can associate to φ the homomorphism $\psi: R / \ker(\varphi) \rightarrow S$ with $\psi([r]_{\ker(\varphi)}) = \varphi(r)$. By construction ψ is injective.